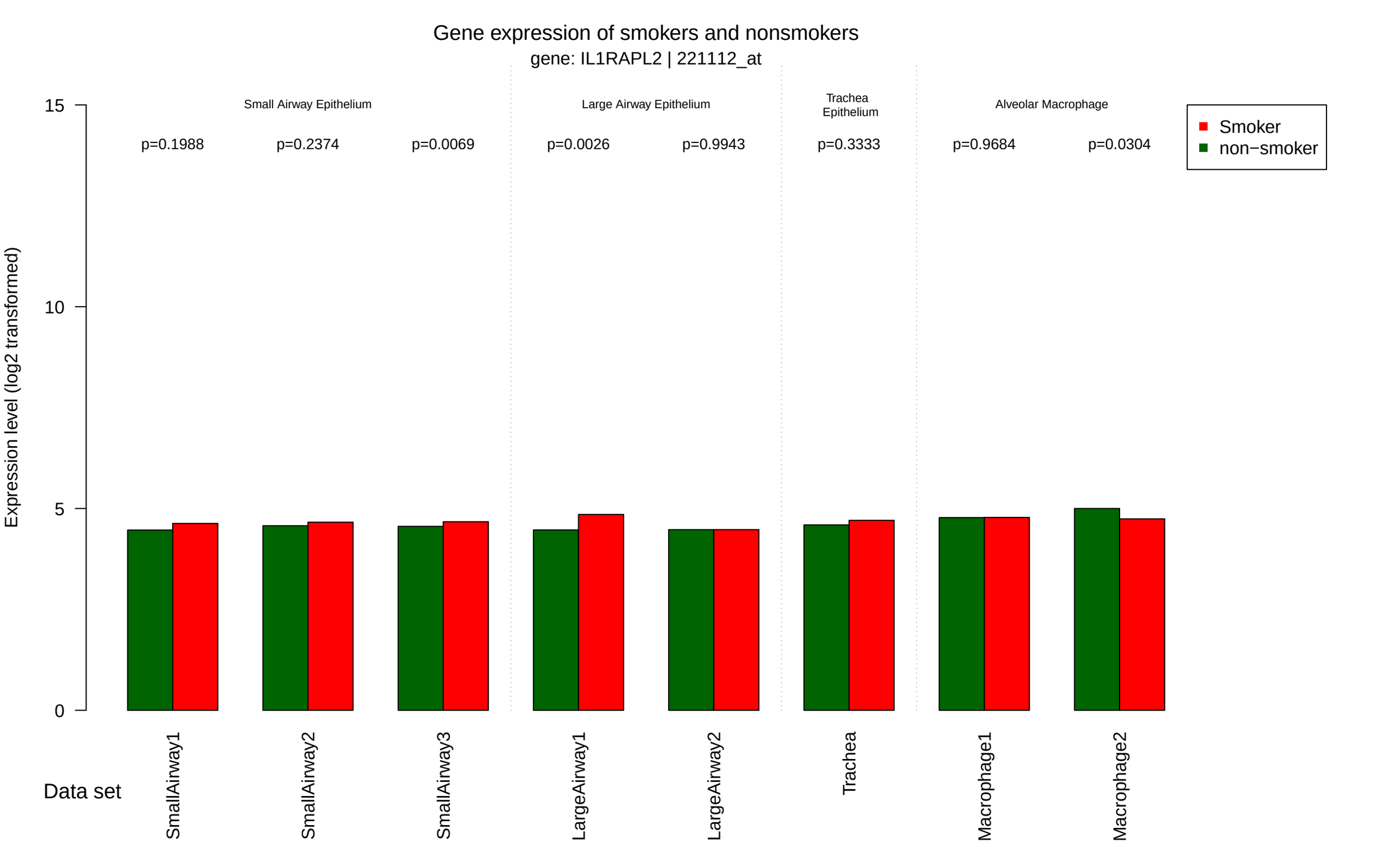
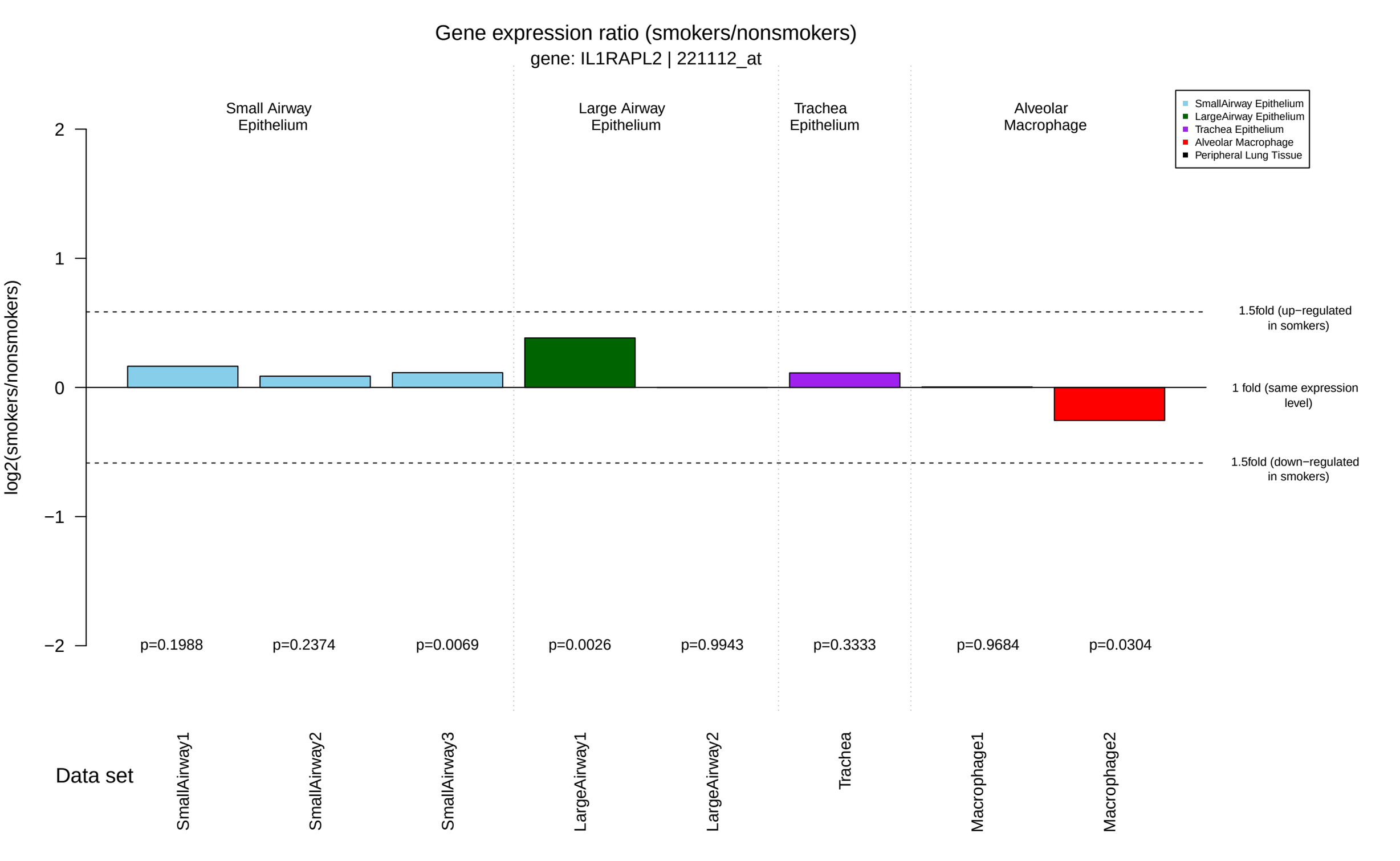
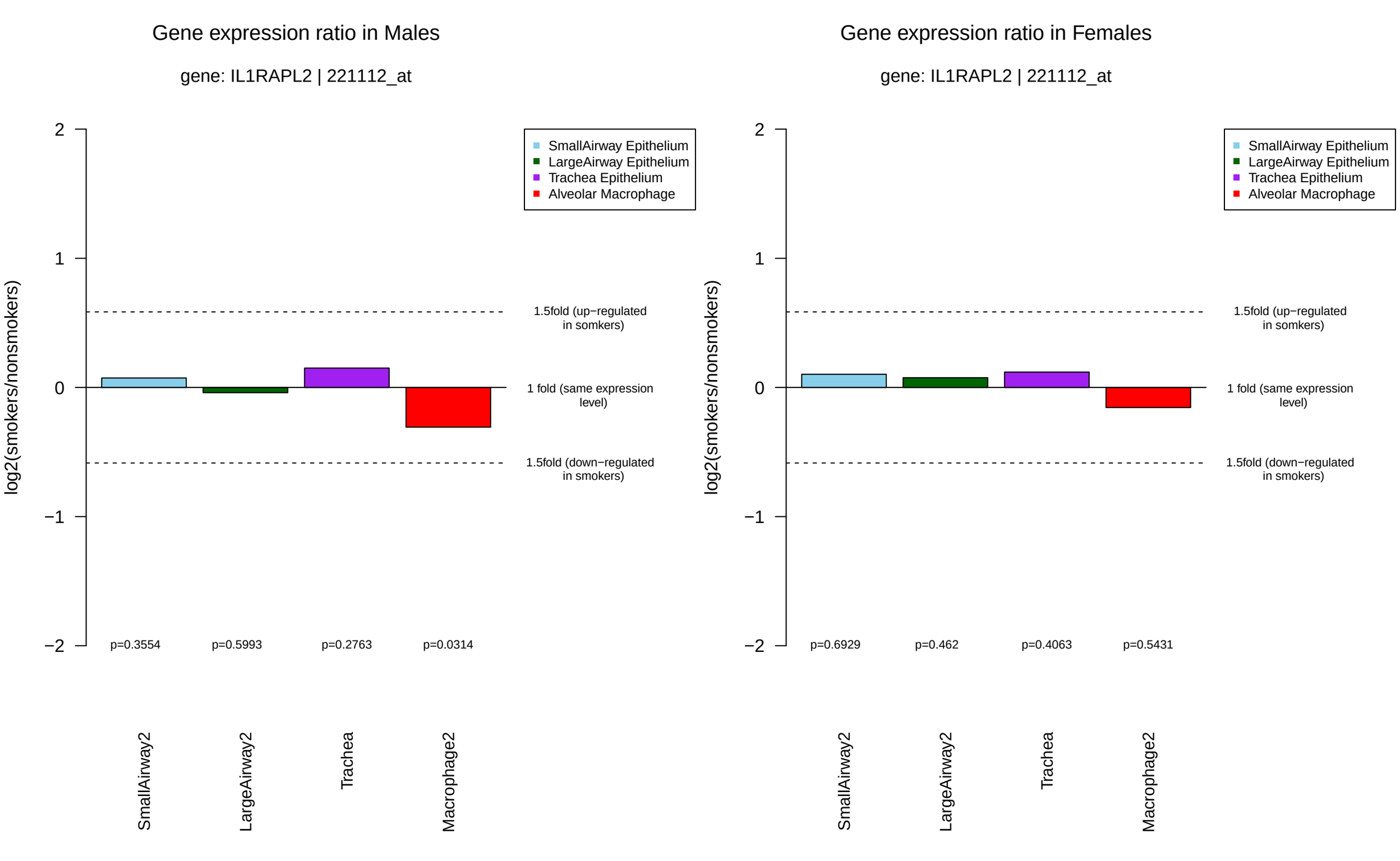
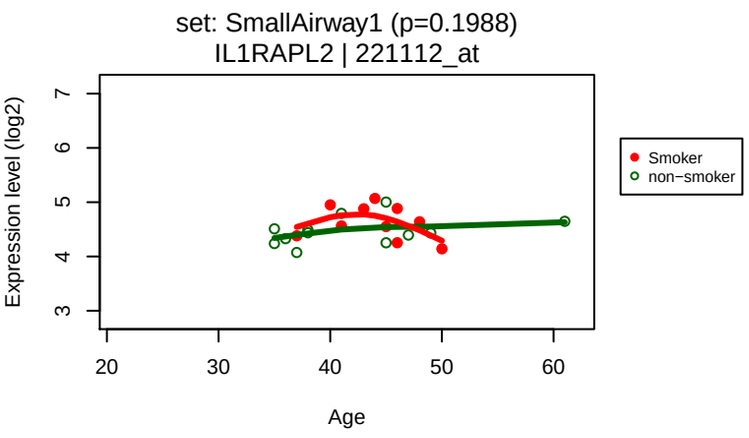
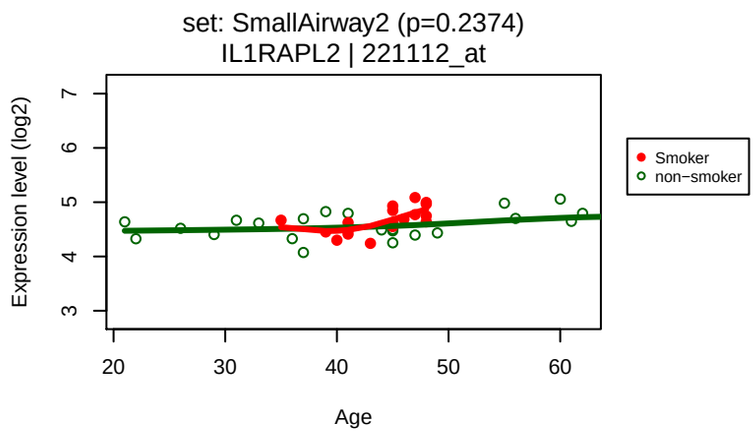
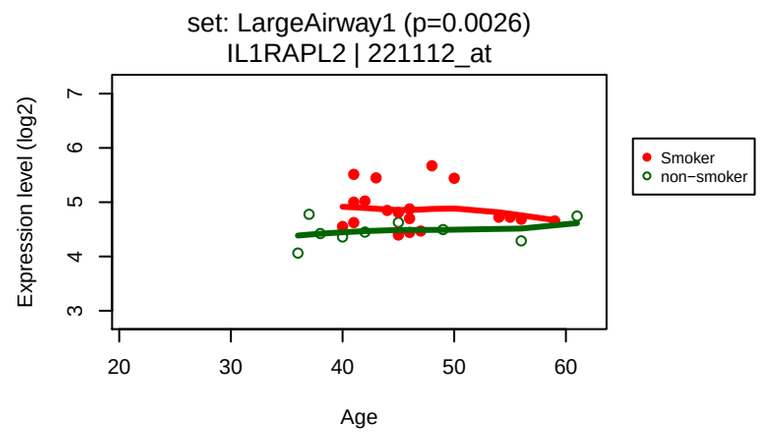

Gene expression of smokers and nonsmokers
gene: IL1RAPL2 | 221112_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.1988
p=0.2374
p=0.0069
p=0.0026
p=0.9943
p=0.3333
p=0.9684
p=0.0304
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

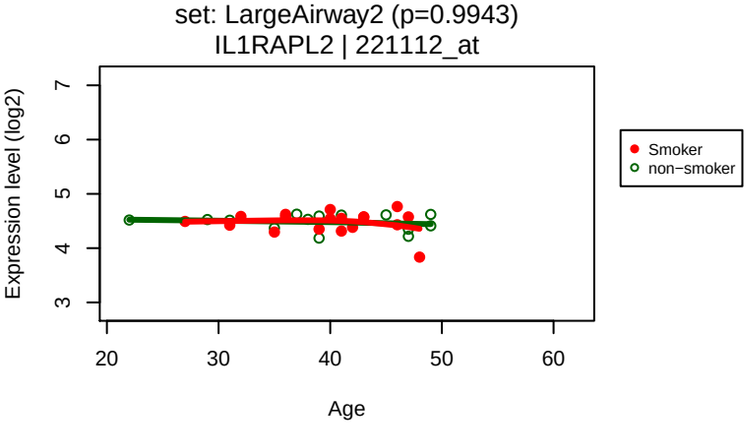

Gene expression ratio (smokers/nonsmokers)
gene: IL1RAPL2 | 221112_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.1988
p=0.2374
p=0.0069
p=0.0026
p=0.9943
p=0.3333
p=0.9684
p=0.0304
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

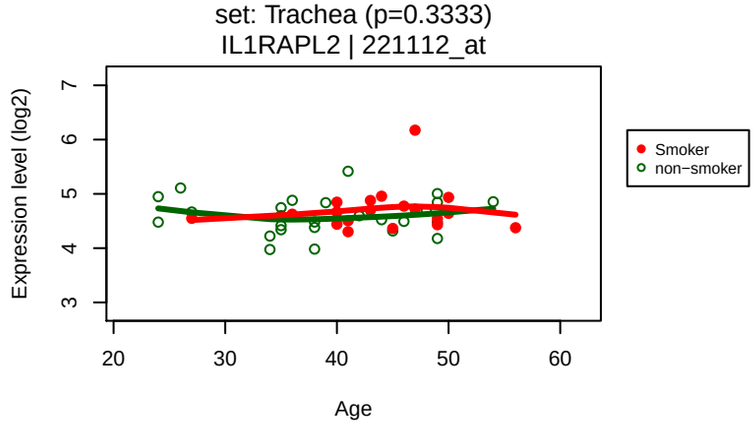

Gene expression ratio in Males
gene: IL1RAPL2 | 221112_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.3554
p=0.5993
p=0.2763
p=0.0314
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: IL1RAPL2 | 221112_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6929
p=0.462
p=0.4063
p=0.5431
SmallAirway2
LargeAirway2
Trachea
Macrophage2

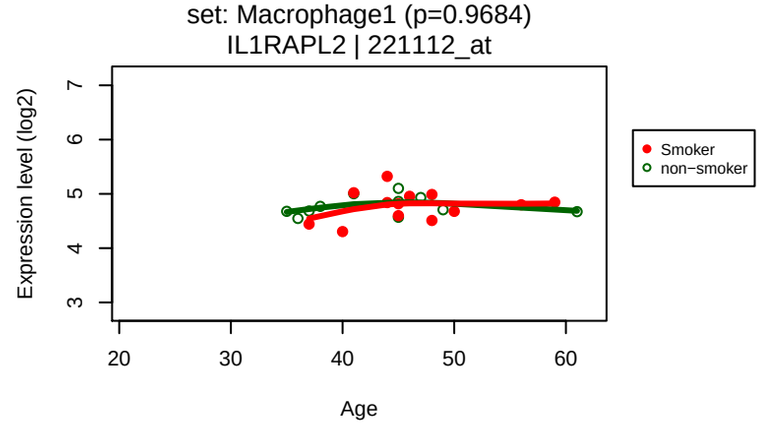

set: SmallAirway1 (p=0.1988)
IL1RAPL2 | 221112_at
Expression level (log2)
Age
Smoker
non−smoker

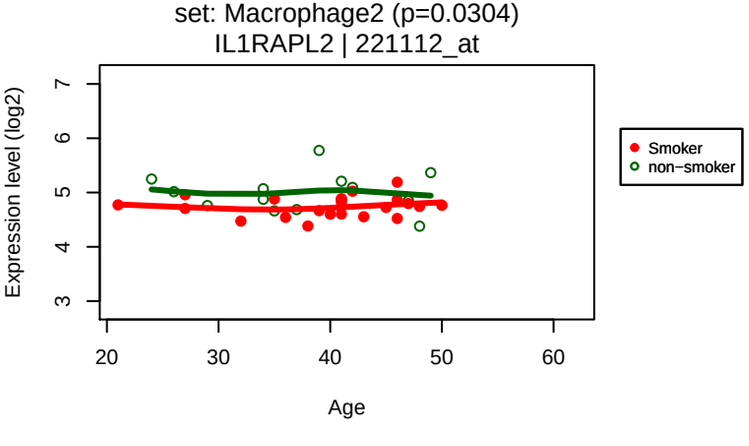

set: SmallAirway2 (p=0.2374)
IL1RAPL2 | 221112_at
Expression level (log2)
Age
Smoker
non−smoker

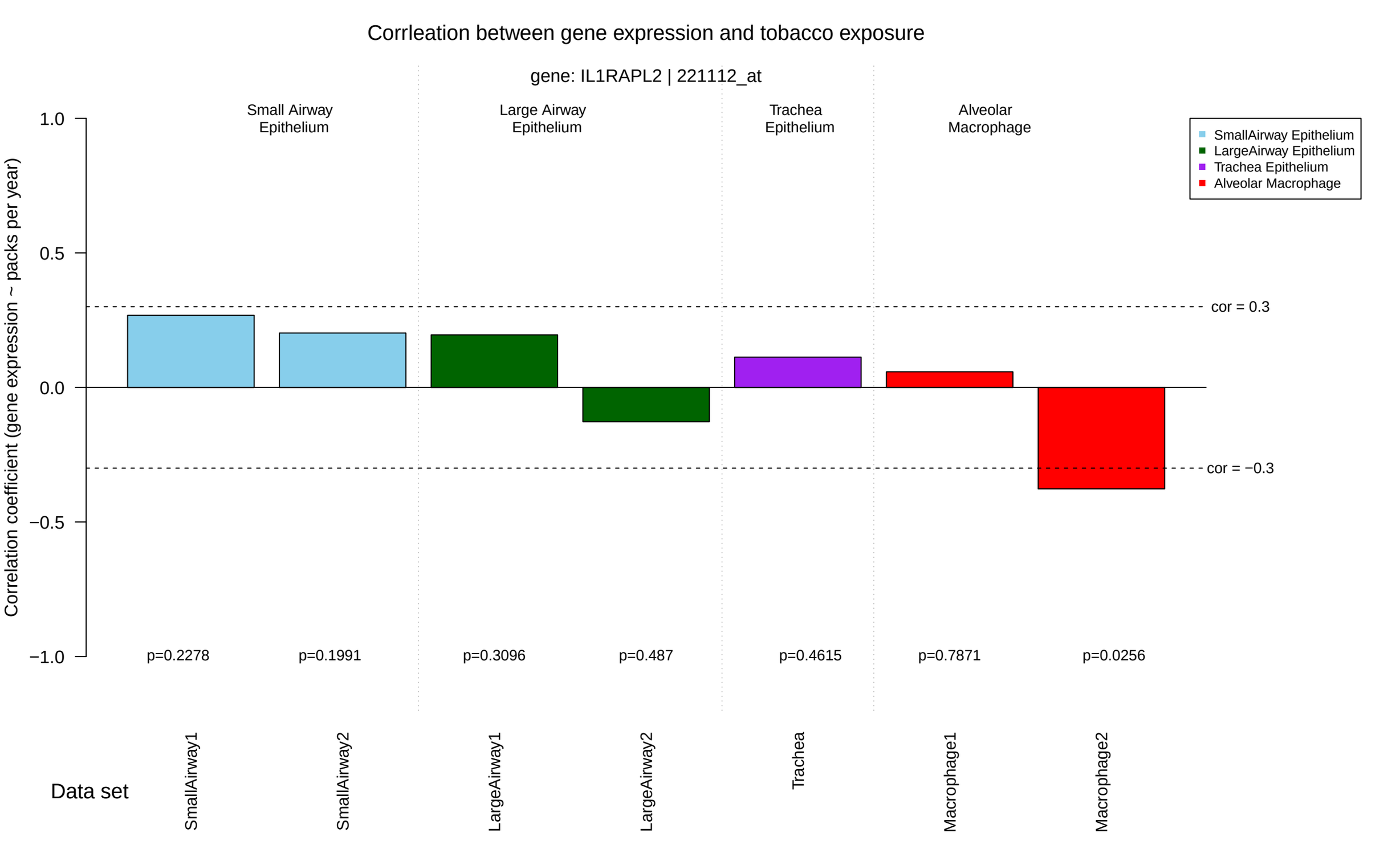

set: LargeAirway1 (p=0.0026)
IL1RAPL2 | 221112_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.9943)
IL1RAPL2 | 221112_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.3333)
IL1RAPL2 | 221112_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9684)
IL1RAPL2 | 221112_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0304)
IL1RAPL2 | 221112_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: IL1RAPL2 | 221112_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.2278
p=0.1991
p=0.3096
p=0.487
p=0.4615
p=0.7871
p=0.0256
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2